

# Introduction to Computer Networks

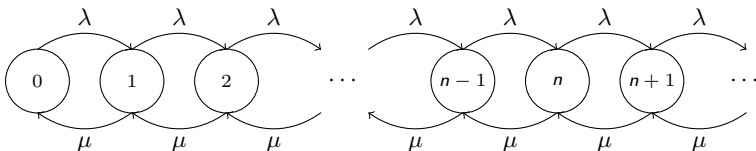
## Recitation 03: Queueing theory

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# Today's recitation: Queueing theory

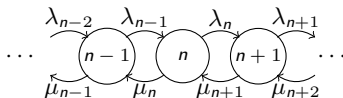
- Basic queueing theory
- Birth-death process
- M/M/1 and other examples
- Multiple access

- Started by Erlang in 1910's
- Big idea: model the arrival times and service times as random processes
  - $\lambda$ : Average time between arrivals  $\Rightarrow$  Probability of an enqueue
  - $\mu$ : Average service time  $\Rightarrow$  Probability of a dequeue



- A Poisson process models:
  - Random independent arrivals
  - Average rate  $\lambda$
  - Stationary increments:  $A(t+s) \sim A(t) + A(s)$ .
- Arrivals after  $t$  time units  $A(t) \sim \text{Poisson}(\lambda t)$ :  $\Pr[A(t) = n] = \frac{(\lambda t)^n}{n!} e^{-\lambda t}$
- Time between arrivals  $T_i \sim \text{Exp}(\lambda)$ :  $\Pr[T_i \leq t] = 1 - e^{-\lambda t}$

- Markov chain:
  - States  $\{0, 1, 2, \dots\}$ , starting at 0
  - Transition probabilities:  $i \rightarrow j$  with probability  $p_{i,j}$
- Birth-death process: only transitions between  $i \rightarrow i+1$  and  $i+1 \rightarrow i$ .



- Steady state
  - $N_t$  : number of clients at time  $t$
  - $p_n = \Pr[N_\infty = n]$
  - Starting with any distribution, after many iterations, the distribution converges to  $p$
  - For birth-death process:

$$p_n = p_0 \cdot \prod_{i=1}^n \frac{\lambda_{i-1}}{\mu_i}$$

## M/M/1 queue

- Markov arrivals ( $\lambda$ ), Markov service times ( $\mu$ ), single server
- Steady state:  $p_n = (1 - \rho) \cdot \rho^n$  where  $\rho = \lambda/\mu$ 
  - Initial:  $p_0 = (1 - \lambda) \cdot p_0 + \mu \cdot p_1$
  - Inductive:  $p_n = \lambda \cdot p_{n-1} + (1 - \lambda - \mu) \cdot p_n + \mu \cdot p_{n+1}$
  - $\sum_{n \geq 0} p_n = 1$

## M/M/1/k

- Queue size at most  $k$
- Steady state:  $p_n = \frac{1-\rho}{1-\rho^{k+1}} \cdot \rho^n$  for  $\rho = \lambda/\mu$
- Only changes the normalization

## M/M/c

- $c$  servers,  $\mu_i = \max(n, c) \cdot \mu$
- Steady state:  $p_n = \begin{cases} p_0 \cdot (c\rho)^n / n! & \text{for } n \leq c \\ p_0 \cdot \rho^n \cdot c^c / c! & \text{for } n \geq c \end{cases}$ , where  $\rho = \lambda/(c\mu)$

## Little's Law

$$N = \lambda T$$

- $N$  is the average number of clients:  $N = \sum_{n \geq 0} n \cdot p_n = \text{Mean}(p)$
- $T$  is the average time spent on the system

## M/G/1

- Service time is a random variable  $S$
- Average number of clients  $N = \frac{\lambda^2 \mathbb{E}[S^2]}{2(1 - \lambda \mathbb{E}[S])}$
- M/D/1
  - Deterministic service time  $D$  ( $\mu = 1/D$ )
  - $N_k$ : number of clients at  $k$ -th departure
  - $N_{k+1} = \max(N_k - 1, 0) + \text{Poisson}(\lambda D)$
  - $p_n = \Pr[N_\infty = n]$
  - $p_n = p_0 \cdot \Pr[\text{Poisson}(\lambda D) = n] + \sum_{i=1}^{n+1} p_i \cdot \Pr[\text{Poisson}(\lambda D) = n + 1 - i]$

- Many clients try to send to the same server
- They may collide in the time slot they try to send
  - If so, both messages fail; must retry
- Probabilistic analysis: Randomly choose a time slot to send

$$\begin{aligned}\mathbb{E}[\text{Time}] &= (\mathbb{E}[\text{Waiting}] + \mathbb{E}[\text{Time}]) \cdot \Pr[\text{Collision}] \\ &\quad + \mathbb{E}[\text{Time}|\text{No Collision}] \cdot \Pr[\text{No Collision}]\end{aligned}$$

- Examples with two senders:
  - Choose uniformly from  $\{1, 2, 3\}$  to send. Retry at the 3 slots passed.

$$\mathbb{E}[\mathcal{T}] = (3 + \mathbb{E}[\mathcal{T}]) \cdot \frac{3}{9} + \left(2 \cdot \frac{2}{6} + 3 \cdot \frac{4}{6}\right) \cdot \frac{6}{9}$$

- Choose uniformly from  $\{1, 2, 3\}$  to send. Retry immediately after.

$$\mathbb{E}[\mathcal{T}] = \left( \left(1 \cdot \frac{1}{3} + 2 \cdot \frac{1}{3} + 3 \cdot \frac{1}{3}\right) + \mathbb{E}[\mathcal{T}] \right) \cdot \frac{3}{9} + \left(2 \cdot \frac{2}{6} + 3 \cdot \frac{4}{6}\right) \cdot \frac{6}{9}$$

# Questions?